Q1 - 2024 (04 Apr Shift 2)

Let $S=\left\{\sin^2 2\theta:\left(\sin^4\theta+\cos^4\theta\right)x^2+(\sin 2\theta)x+\left(\sin^6\theta+\cos^6\theta\right)=0\text{ has real roots }\right\}$. If α and β be the smallest and largest elements of the set S, respectively, then $3\left((\alpha-2)^2+(\beta-1)^2\right)$ equals _________

Q2 - 2024 (05 Apr Shift 1)

Suppose $\theta\epsilon\left[0,\frac{\pi}{4}\right]$ is a solution of $4\cos\theta-3\sin\theta=1$. Then $\cos\theta$ is equal to :

(1) $\frac{4}{(3\sqrt{6}+2)}$

(2) $\frac{6+\sqrt{6}}{(3\sqrt{6}+2)}$

(3) $\frac{4}{(3\sqrt{6}-2)}$

(4) $\frac{6-\sqrt{6}}{(3\sqrt{6}-2)}$

Q3 - 2024 (05 Apr Shift 2)

The number of solutions of $\sin^2 x+\left(2+2x-x^2\right)\sin x-3(x-1)^2=0$, where $-\pi\leq x\leq\pi$, is________

Answer Key

Q1 (4) **Q2** (3) **Q3** (2)

Q1

$$D=(\sin 2\theta)^2-4\left(1-\frac{\sin^2 2\theta}{2}\right)\left(1-\frac{3}{4}\sin^2 2\theta\right)$$

$$=(\sin 2\theta)^2-4\left(1-\frac{5}{4}\sin^2 2\theta+\frac{3}{8}\sin^4 2\theta\right)$$

$$D=-\frac{3}{2}\sin^4 2\theta+6\sin^2 2\theta-4>0$$

$$3\sin^4 2\theta-12\sin^2 2\theta+8<0$$

$$\sin^2 2\theta=\frac{12\pm\sqrt{12^2-12.8}}{6}=\frac{12\pm 4\sqrt{3}}{6}=\frac{6\pm 2\sqrt{3}}{3}$$

$$\sin^2 2\theta=2\pm\frac{2}{\sqrt{3}},\ \text{but}\ \sin^2 2\theta\in[0,1]$$

$$\therefore \alpha=2-\frac{2}{\sqrt{3}},\beta=1\rightarrow(\alpha-2)^2=\frac{4}{3},(\beta-1)^2=0$$

$$3(\alpha-2)^2+(\beta-1)^2=4$$

Q2

$$4\left(\frac{1-\tan^2\theta/2}{1+\tan^2\theta/2}\right)-3\left(\frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}\right)=1$$

let $\tan\frac{\theta}{2}=\mathrm{t}$

$$\frac{4-4t^2-6t}{1+t^2}=1$$

$$4-4t^2-6t=1+t^2$$
$$\Rightarrow 5t^2+6t-3=0$$

$$\Rightarrow t=\frac{-6\pm\sqrt{36-4(5)(-3)}}{2(5)}$$

$$=\frac{-6\pm\sqrt{96}}{10}$$

$$=\frac{-6\pm4\sqrt{6}}{10}$$

$$t=\frac{-3+2\sqrt{6}}{5}$$

$$\cos\theta=\frac{1-t^2}{1+t^2}=\frac{1-\left(\frac{2\sqrt{6}-3}{5}\right)^2}{1+\left(\frac{2\sqrt{6}-3}{5}\right)^2}=\frac{1-\left(\frac{24+9-12\sqrt{6}}{25}\right)}{1+\left(\frac{24+9-12\sqrt{6}}{25}\right)}$$

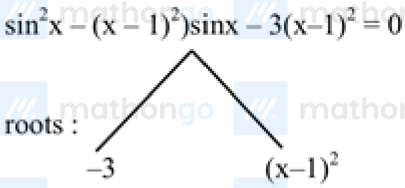
$$=\frac{25-33+12\sqrt{6}}{25+33-12\sqrt{6}}=\frac{12\sqrt{6}-8}{58-12\sqrt{6}}=\frac{6\sqrt{6}-4}{29-6\sqrt{6}}\times\frac{29+6\sqrt{6}}{29+6\sqrt{6}}$$

$$=\frac{100+150\sqrt{6}}{625}=\frac{4+6\sqrt{6}}{25}\times\frac{4-6\sqrt{6}}{4-6\sqrt{6}}$$

$$=\frac{-200}{25(4-6\sqrt{6})}=\frac{-8}{4-6\sqrt{6}}=\frac{4}{3\sqrt{6}-2}$$

Q3

$$\sin^2 x-\left(x^2-2x-2\right)\sin x-3(x-1)^2=0$$



$\sin x=-3(\text{ reject }) \text{ or } (x-1)^2$

$\sin x=(x-1)^2$